Final Project

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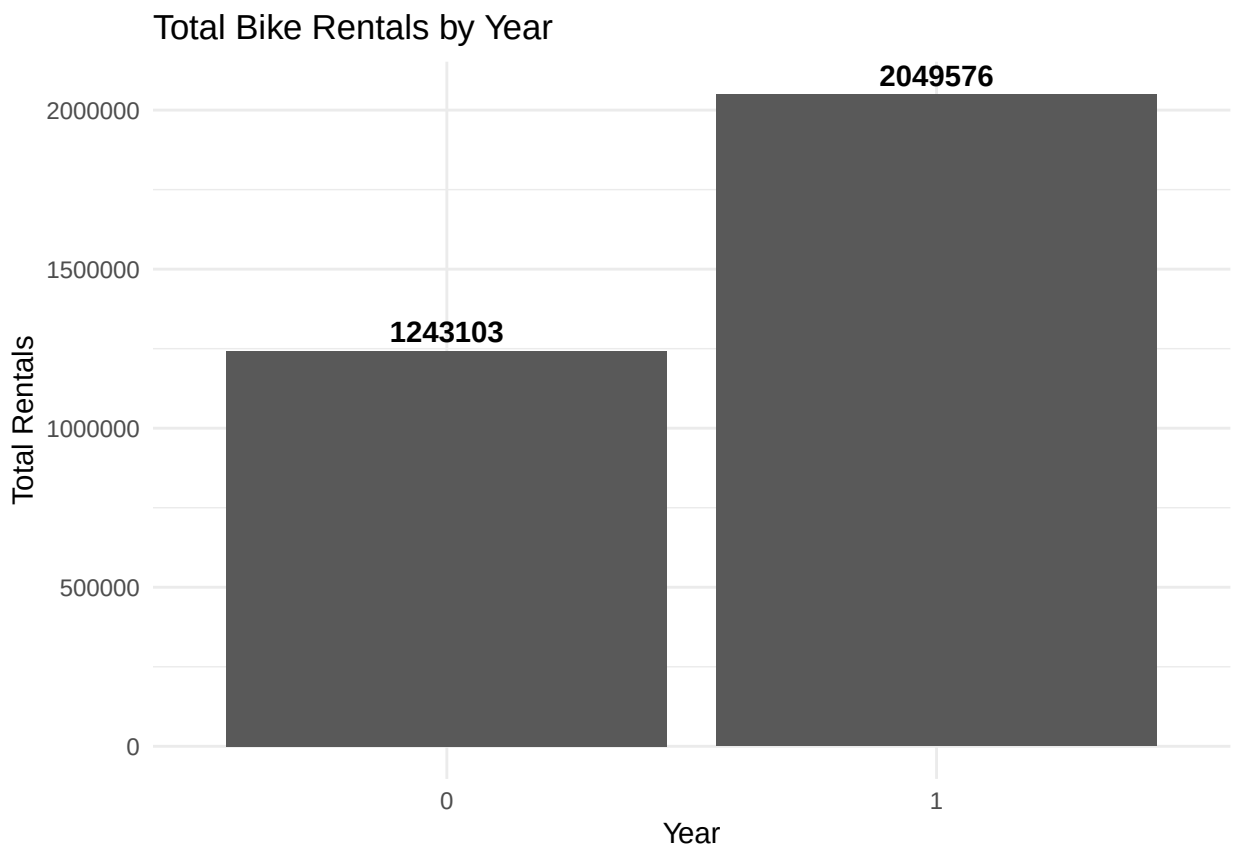
Part 1

1 - Read Data and EDA

```
# Read Data
bike_raw <- read.csv("bike_sharing_daily.csv")

# Re-assign values of variables
bike_raw <- bike_raw %>%
  mutate(
    season     = factor(season,     levels = 1:4,
                        labels = c("Spring", "Summer", "Fall", "Winter")),
    mnth       = factor(mnth,       levels = 1:12),
    holiday    = factor(holiday,    levels = c(0,1), labels = c("No", "Yes")),
    weekday    = factor(weekday,    levels = 0:6,
                        labels = c("Sun","Mon","Tue","Wed","Thu","Fri","Sat")),
    workingday = factor(workingday, levels = c(0,1), labels = c("No", "Yes")),
    weathersit = factor(weathersit, levels = 1:4,
                        labels = c("Clear", "Mist/Cloudy", "Light Rain/Snow", "Heavy Rain/Snow"))
  )

# Filter for 2011
bike_2011 <- bike_raw %>% filter(yr == 0)
```

Look at bike rentals share over the years

```
# Bar plot of total rentals by year
bike_raw %>%
  group_by(yr) %>%
  summarise(total_rentals = sum(cnt)) %>%
  ggplot(aes(x = factor(yr), y = total_rentals)) +
  geom_col() +
  geom_text(aes(label = total_rentals), vjust = -0.4, fontface = "bold") +
  labs(title = "Total Bike Rentals by Year",
       x = "Year", y = "Total Rentals") +
  theme_minimal()
```



Bike rentals increased significantly from 2011 to 2012, with total rentals in 2012 being almost double that of 2011. This suggests that there may be underlying trends or changes in the bike sharing system between the two years that could affect model performance when training on 2011 data and testing on 2012 data.

Categorical Variables

Season, month, holiday, weekday, and workingday are standard characteristics within a year. There is no reason to expect significant variation in their distributions. Thus, we will not print out summary tables and plots for these variables.

```
table(bike_2011$weathersit)
```

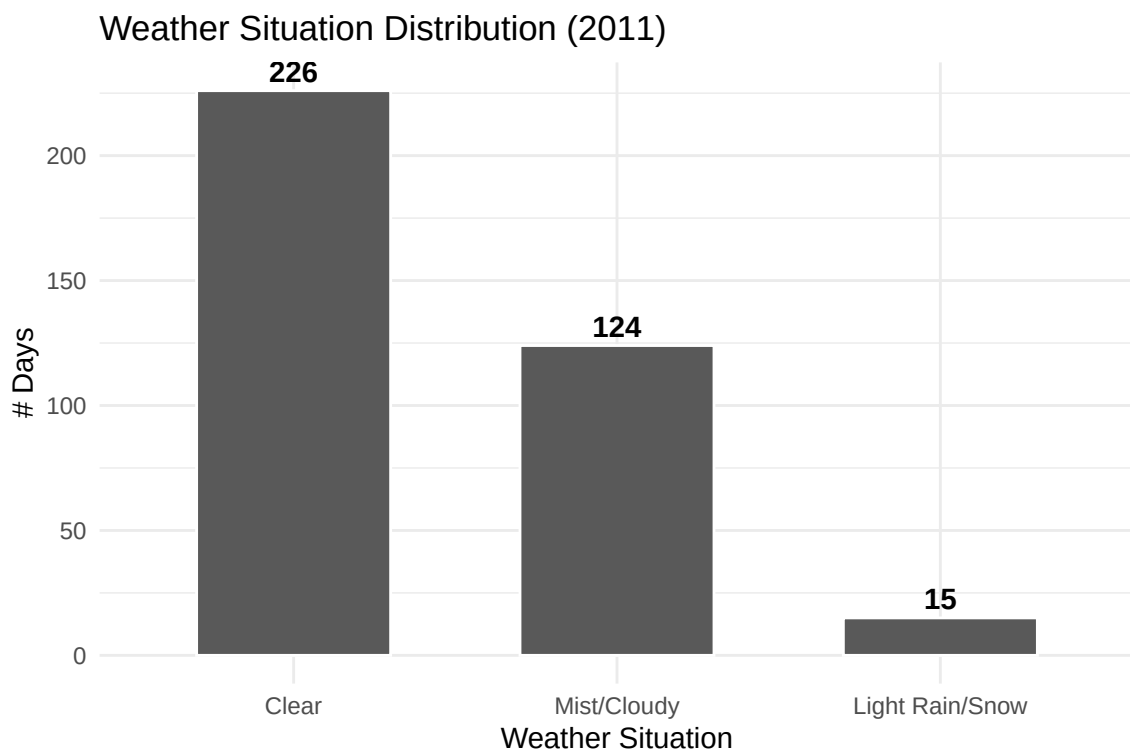
Weather Situation

```
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##             226             124              15               0
```

```
prop.table(table(bike_2011$weathersit))
```

```
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##      0.61917808      0.33972603      0.04109589      0.00000000
```

```r
bike_2011 %>%
  count(weathersit) %>%
  ggplot(aes(x = weathersit, y = n)) +
  geom_col(width = 0.6, color = "white") +
  geom_text(aes(label = n), vjust = -0.4, fontface = "bold") +
  labs(title = "Weather Situation Distribution (2011)",
       x = "Weather Situation", y = "# Days") +
  theme_minimal()
```



Continuous Variables

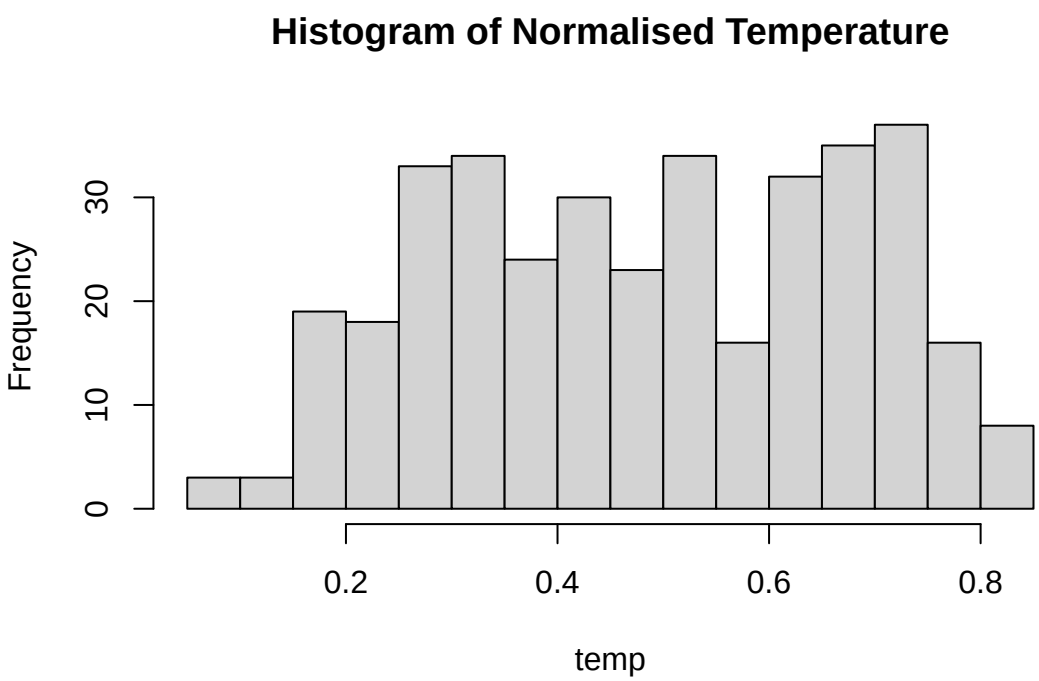
```r
cont_vars <- bike_2011 %>% select(temp, atemp, hum, windspeed, casual, registered, cnt)

# Numerical summary for all continuous variables
summary(cont_vars)
```

```
##       temp             atemp              hum           windspeed
##  Min.   :0.05913   Min.   :0.07907   Min.   :0.0000   Min.   :0.02239
##  1st Qu.:0.32500   1st Qu.:0.32195   1st Qu.:0.5383   1st Qu.:0.13558
##  Median :0.47917   Median :0.47285   Median :0.6475   Median :0.18690
##  Mean   :0.48666   Mean   :0.46684   Mean   :0.6437   Mean   :0.19140
##  3rd Qu.:0.65667   3rd Qu.:0.61238   3rd Qu.:0.7421   3rd Qu.:0.23508
##  Max.   :0.84917   Max.   :0.84090   Max.   :0.9725   Max.   :0.50746
##      casual         registered        cnt
##  Min.   :   9.0   Min.   : 416   Min.   : 431
##  1st Qu.: 222.0   1st Qu.:1730   1st Qu.:2132
```

```
##  Median : 614.0   Median :2915   Median :3740
##  Mean   : 677.4   Mean   :2728   Mean   :3406
##  3rd Qu.: 871.0   3rd Qu.:3632   3rd Qu.:4586
##  Max.   :3065.0   Max.   :4614   Max.   :6043
```

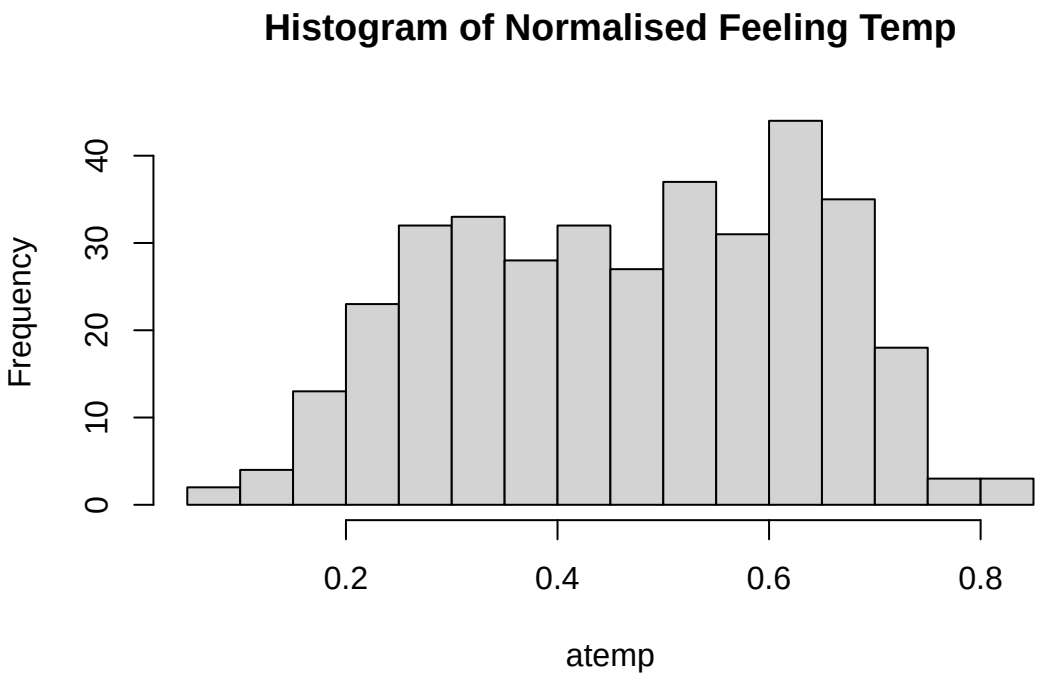
```r
hist(bike_2011$temp, main = "Histogram of Normalised Temperature",
     xlab = "temp", breaks = 20)
```

Temperature (`temp`)



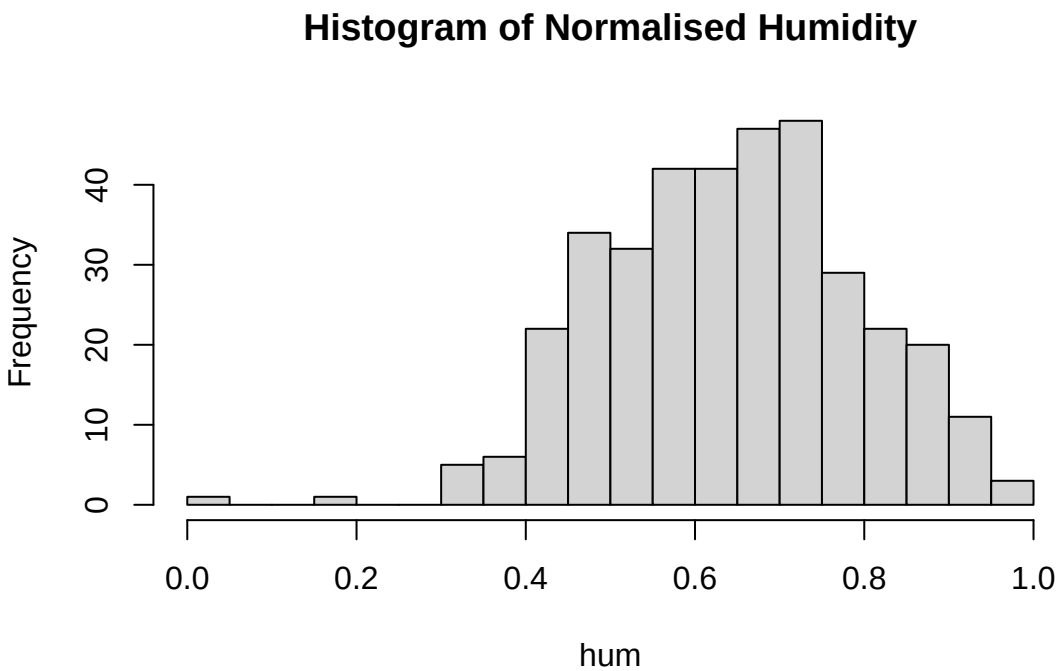

```r
hist(bike_2011$atemp, main = "Histogram of Normalised Feeling Temp",
     xlab = "atemp", breaks = 20)
```

Feeling Temperature (`atemp`)



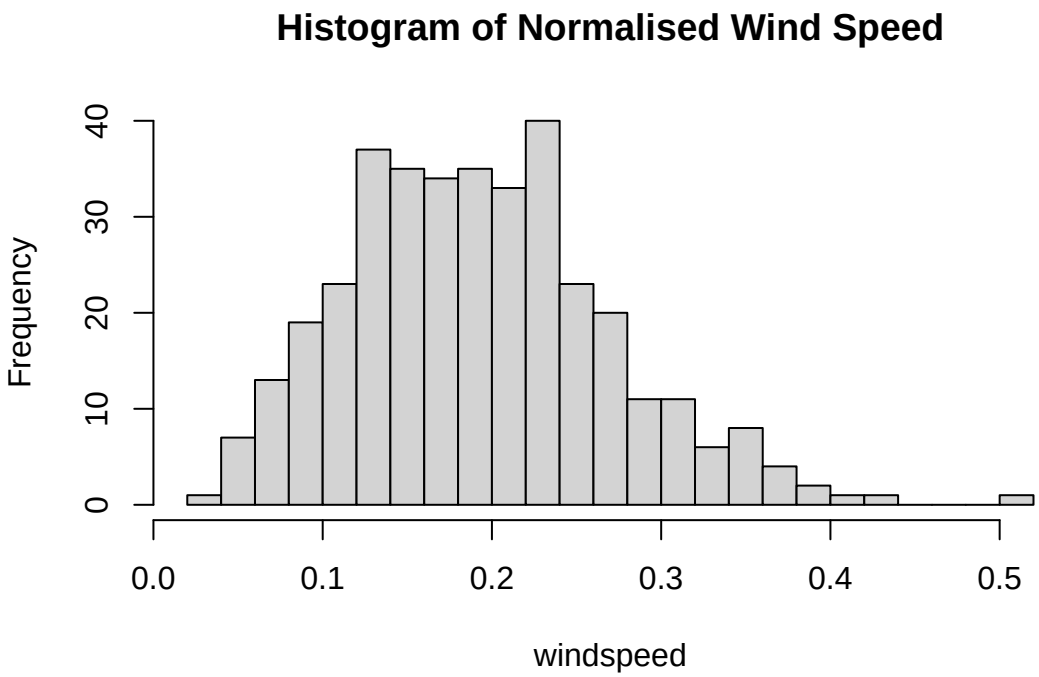

```r
hist(bike_2011$hum, main = "Histogram of Normalised Humidity",
     xlab = "hum", breaks = 20)
```

Humidity (`hum`)



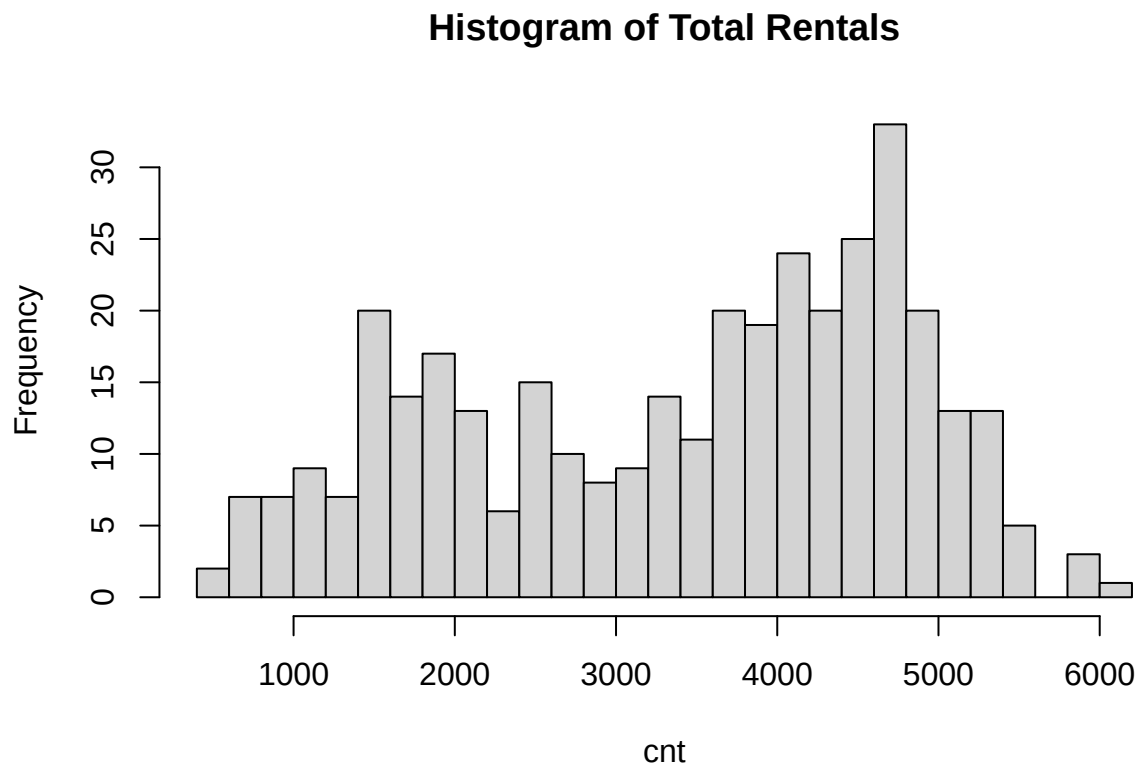

```
hist(bike_2011$windspeed, main = "Histogram of Normalised Wind Speed",
     xlab = "windspeed", breaks = 20)
```

Wind Speed (`windspeed`)



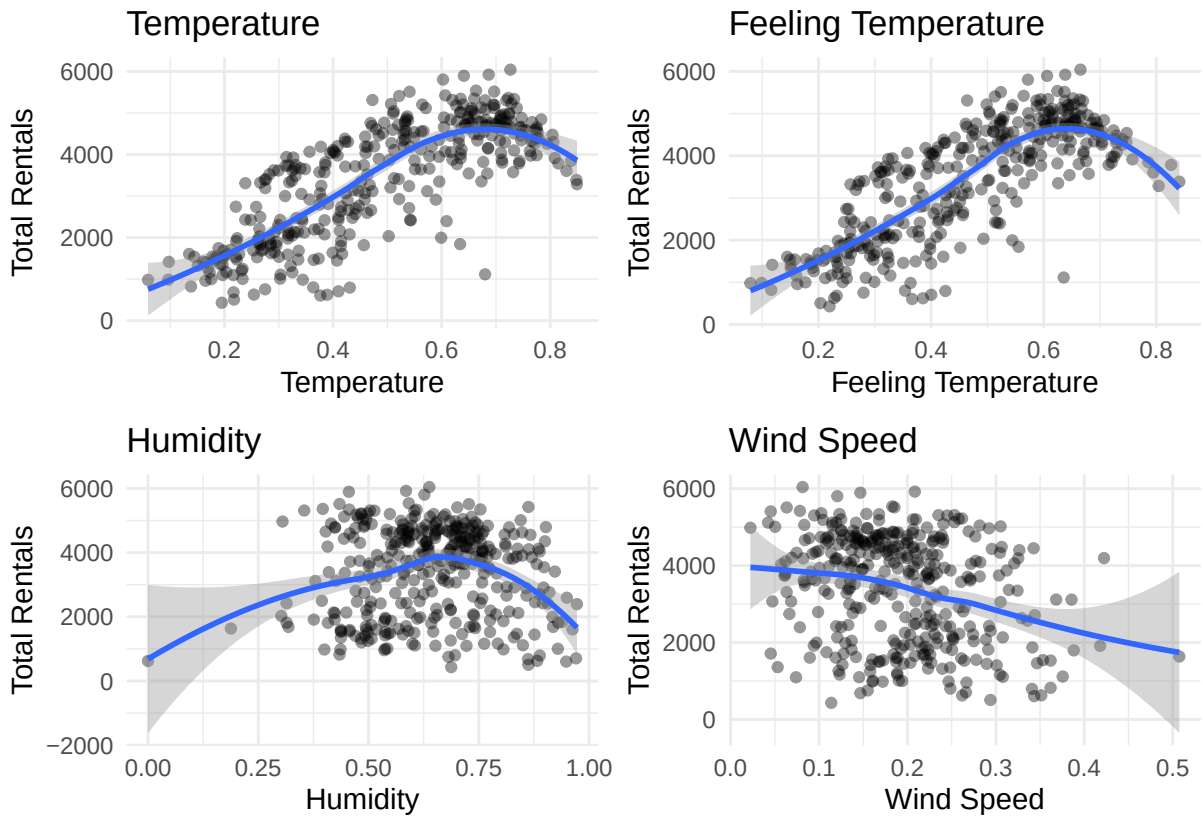

```r
hist(bike_2011$cnt, main = "Histogram of Total Rentals",
     xlab = "cnt", breaks = 25)
```

Total Rentals (`cnt`)



Continuous variables vs cnt

```
p1 <- ggplot(bike_2011, aes(x = temp, y = cnt)) + geom_point(alpha = 0.4) + geom_smooth() +
  labs(title = "Temperature", x = "Temperature", y = "Total Rentals") +
  theme_minimal()
p2 <- ggplot(bike_2011, aes(x = atemp, y = cnt)) + geom_point(alpha = 0.4) + geom_smooth() +
  labs(title = "Feeling Temperature", x = "Feeling Temperature", y = "Total Rentals") +
  theme_minimal()
p3 <- ggplot(bike_2011, aes(x = hum, y = cnt)) + geom_point(alpha = 0.4) + geom_smooth() +
  labs(title = "Humidity", x = "Humidity", y = "Total Rentals") +
  theme_minimal()
p4 <- ggplot(bike_2011, aes(x = windspeed, y = cnt)) + geom_point(alpha = 0.4) + geom_smooth() +
  labs(title = "Wind Speed", x = "Wind Speed", y = "Total Rentals") +
  theme_minimal()
p1 + p2 + p3 + p4 + plot_layout(ncol = 2)
```



Correlation matrix for continuous variables

```
# correlation matrix
cont_vars <- bike_2011 %>% select(temp, atemp, hum, windspeed, casual, registered, cnt)
cor_matrix <- cor(cont_vars)
print(cor_matrix)
```

```
##                  temp      atemp          hum  windspeed      casual
## temp        1.0000000  0.9964576  0.145776184 -0.1142002  0.58103786
## atemp       0.9964576  1.0000000  0.155811515 -0.1365438  0.58115314
## hum         0.1457762  0.1558115  1.000000000 -0.2157180 -0.03229046
## windspeed  -0.1142002 -0.1365438 -0.215718023  1.0000000 -0.19051690
## casual      0.5810379  0.5811531 -0.032290458 -0.1905169  1.00000000
## registered  0.6981358  0.7033810  0.019412295 -0.2615905  0.39654650
## cnt         0.7712142  0.7752937  0.001898085 -0.2780000  0.70835873
##             registered          cnt
## temp        0.69813575  0.771214198
## atemp       0.70338097  0.775293710
## hum         0.01941229  0.001898085
## windspeed  -0.26159050 -0.277999968
## casual      0.39654650  0.708358731
## registered  1.00000000  0.928880205
## cnt         0.92888020  1.000000000
```

Temp and atemp are highly correlated. In a model with multiple predictors, we would likely exclude one of them to avoid multicollinearity.

2 - Fit linear regression models predicting `cnt` using each of the variables separately.

cnt ~ season

```r
model_season <- lm(cnt ~ season, data = bike_2011)
summary(model_season)
```

```
##
## Call:
## lm(formula = cnt ~ season, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3349.4  -504.7    96.8   593.6  2029.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1666.67      95.43   17.46   <2e-16 ***
## seasonSummer  2108.51     134.23   15.71   <2e-16 ***
## seasonFall    2797.70     133.52   20.95   <2e-16 ***
## seasonWinter  1997.79     135.34   14.76   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 905.4 on 361 degrees of freedom
## Multiple R-squared:  0.5724, Adjusted R-squared:  0.5688
## F-statistic: 161.1 on 3 and 361 DF,  p-value: < 2.2e-16
```

The model explains about 57% of the variance in total rentals. The coefficients indicate that, compared to Spring, summer has 2108 more rentals, fall has 2797 more rentals, and winter has 1997 more rentals on average. Overall, season is a good predictor of total rentals.

cnt ~ mnth

```r
model_mnth <- lm(cnt ~ mnth, data = bike_2011)
summary(model_mnth)
```

```
##
## Call:
## lm(formula = cnt ~ mnth, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3357.2  -315.2    66.0   479.7  2149.7
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1231.9      137.6   8.950  < 2e-16 ***
## mnth2          490.1      199.8   2.453   0.0147 *
## mnth3          834.1      194.7   4.285 2.36e-05 ***
## mnth4         1930.4      196.3   9.835  < 2e-16 ***
## mnth5         3149.4      194.7  16.179  < 2e-16 ***
## mnth6         3551.8      196.3  18.096  < 2e-16 ***
## mnth7         3327.5      194.7  17.094  < 2e-16 ***
## mnth8         3177.5      194.7  16.323  < 2e-16 ***
## mnth9         3015.4      196.3  15.363  < 2e-16 ***
## mnth10        2752.3      194.7  14.139  < 2e-16 ***
## mnth11        2173.7      196.3  11.075  < 2e-16 ***
## mnth12        1585.0      194.7   8.142 6.75e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 766.4 on 353 degrees of freedom
## Multiple R-squared:  0.7004, Adjusted R-squared:  0.691
## F-statistic: 75.01 on 11 and 353 DF,  p-value: < 2.2e-16
```

The model has a better R2 than season, explaining about 70% of the variance in total rentals. Due to the strong corelation between month and season, the coefficients for month show a clear seasonal pattern. Rentals peak in the summer months (6, 7, 8).

cnt ~ holiday

```r
model_holiday <- lm(cnt ~ holiday, data = bike_2011)
summary(model_holiday)
```

```
##
## Call:
## lm(formula = cnt ~ holiday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2986.1 -1285.1   329.9  1168.9  3040.8
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3417.13      73.19  46.689   <2e-16 ***
## holidayYes   -414.93     442.17  -0.938    0.349
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1379 on 363 degrees of freedom
## Multiple R-squared:  0.00242,    Adjusted R-squared:  -0.0003282
## F-statistic: 0.8806 on 1 and 363 DF,  p-value: 0.3487
```

The model has a very low R2 of 0.002, indicating that holiday status is not a good predictor of total rentals.

cnt ~ weekday

```r
model_weekday <- lm(cnt ~ weekday, data = bike_2011)
summary(model_weekday)
```

```
##
## Call:
## lm(formula = cnt ~ weekday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2925.8 -1273.1   345.2  1147.7  2577.2
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3405.27     192.49  17.691   <2e-16 ***
## weekdayMon     60.52     272.22   0.222    0.824
## weekdayTue     62.77     272.22   0.231    0.818
## weekdayWed   -152.02     272.22  -0.558    0.577
## weekdayThu    -48.50     272.22  -0.178    0.859
## weekdayFri     94.85     272.22   0.348    0.728
## weekdaySat    -13.89     270.93  -0.051    0.959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1388 on 358 degrees of freedom
## Multiple R-squared:  0.003175,   Adjusted R-squared:  -0.01353
## F-statistic: 0.1901 on 6 and 358 DF,  p-value: 0.9795
```

This model also has a low R2 of 0.003, suggesting that day of the week is a bad predictor of total rentals.

cnt ~ workingday

```r
model_workingday <- lm(cnt ~ workingday, data = bike_2011)
summary(model_workingday)
```

```
##
## Call:
## lm(formula = cnt ~ workingday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2994.1 -1291.1   321.9  1164.9  2679.2
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3363.82     128.72  26.133   <2e-16 ***
## workingdayYes    61.24     155.53   0.394    0.694
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1380 on 363 degrees of freedom
## Multiple R-squared:  0.0004269,  Adjusted R-squared:  -0.002327
## F-statistic: 0.155 on 1 and 363 DF,  p-value: 0.694
```

This model's R2 is worse. At 0.0004, it indicates that whether it's a working day or not has no predictive power for total rentals.

cnt ~ weathersit

```r
model_weathersit <- lm(cnt ~ weathersit, data = bike_2011)
summary(model_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ weathersit, data = bike_2011)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3264  -1103    373   1032   2955
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                3694.99      86.74  42.598  < 2e-16 ***
## weathersitMist/Cloudy      -606.89     145.73  -4.164 3.90e-05 ***
## weathersitLight Rain/Snow -2020.85     347.69  -5.812 1.35e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1304 on 362 degrees of freedom
## Multiple R-squared:  0.1104, Adjusted R-squared:  0.1055
## F-statistic: 22.46 on 2 and 362 DF,  p-value: 6.363e-10
```

This model has an R2 of 0.11. The coefficients show that, compared to clear weather, adverse weather conditions (mist/cloudy, light rain/snow, heavy rain/snow) are associated with fewer rentals. However, the model's overall predictive power is still quite low.

cnt ~ temp

```r
model_temp <- lm(cnt ~ temp, data = bike_2011)
summary(model_temp)
```

```
##
## Call:
## lm(formula = cnt ~ temp, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3375.0  -501.2   -60.3   677.3  1985.7
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    676.4      126.9   5.331 1.72e-07 ***
## temp          5608.3      243.0  23.082  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 878.9 on 363 degrees of freedom
## Multiple R-squared:  0.5948, Adjusted R-squared:  0.5937
## F-statistic: 532.8 on 1 and 363 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.59, indicating that temperature is a strong predictor of total rentals. The positive coefficient suggests that higher temperatures are associated with more rentals.

cnt ~ atemp

```r
model_atemp <- lm(cnt ~ atemp, data = bike_2011)
summary(model_atemp)
```

```
##
## Call:
## lm(formula = cnt ~ atemp, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3359.0  -536.9   -41.5   680.9  1924.1
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    450.1      134.4    3.35 0.000893 ***
## atemp         6331.2      270.7   23.39  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 872 on 363 degrees of freedom
## Multiple R-squared:  0.6011, Adjusted R-squared:    0.6
## F-statistic:   547 on 1 and 363 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.6, slightly better than temp. The positive coefficient indicates that higher feeling temperatures are associated with more rentals.

cnt ~ hum

```r
model_hum <- lm(cnt ~ hum, data = bike_2011)
summary(model_hum)
```

```
##
## Call:
```

```
## lm(formula = cnt ~ hum, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2975.5 -1272.9   333.9  1180.9  2637.3
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3394.44     321.38  10.562   <2e-16 ***
## hum            17.59     486.51   0.036    0.971
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1381 on 363 degrees of freedom
## Multiple R-squared:  3.603e-06,  Adjusted R-squared:  -0.002751
## F-statistic: 0.001308 on 1 and 363 DF,  p-value: 0.9712
```

This model has a very low R2, the p-value for hum is also not significant. This suggests that humidity is not a good predictor of total rentals.

cnt ~ windspeed

```r
model_windspeed <- lm(cnt ~ windspeed, data = bike_2011)
summary(model_windspeed)
```

```
##
## Call:
## lm(formula = cnt ~ windspeed, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3361.4 -1071.1   275.1  1068.4  2601.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4359.9      186.4  23.384  < 2e-16 ***
## windspeed    -4985.0      904.1  -5.514 6.67e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1326 on 363 degrees of freedom
## Multiple R-squared:  0.07728,    Adjusted R-squared:  0.07474
## F-statistic:  30.4 on 1 and 363 DF,  p-value: 6.673e-08
```

This model has a low R2 pf 0.07, indicating that wind speed is a weak predictor of total rentals.

3 - Fit multiple linear regression models with interactions.

Given that season, mnth, weathersit, temp, and atemp are good predictors individually, we will explore interactions between these variables.

cnt ~ temp*weathersit

```r
model_temp_weathersit <- lm(cnt ~ temp * weathersit, data = bike_2011)
summary(model_temp_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ temp * weathersit, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3221.3  -544.7     3.9   625.4  1760.1
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      1052.1      142.1   7.406 9.38e-13 ***
## temp                             5290.6      263.7  20.063  < 2e-16 ***
## weathersitMist/Cloudy            -733.7      254.6  -2.882  0.00419 **
## weathersitLight Rain/Snow       -1279.5      805.0  -1.589  0.11283
## temp:weathersitMist/Cloudy        618.0      499.2   1.238  0.21648
## temp:weathersitLight Rain/Snow   -976.5     1755.4  -0.556  0.57837
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 799.9 on 359 degrees of freedom
## Multiple R-squared:  0.668,  Adjusted R-squared:  0.6634
## F-statistic: 144.5 on 5 and 359 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.67, which is an improvement over the temp-only model. The interaction terms are not significant, suggesting that the effect of temperature on rentals does not differ significantly across weather situations. However, the main effect of temp and cloudy weather remain strong and significant.

cnt ~ temp*season

```r
model_temp_season <- lm(cnt ~ temp * season, data = bike_2011)
summary(model_temp_season)
```

```
##
## Call:
## lm(formula = cnt ~ temp * season, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3344.1  -307.4   120.4   438.9  1922.7
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)          638.3      237.1   2.691  0.00745 **
## temp                3759.8      818.0   4.596 5.97e-06 ***
## seasonSummer        -201.2      388.7  -0.518  0.60507
```

```
## seasonFall          3652.7      783.0   4.665 4.37e-06 ***
## seasonWinter        1349.2      401.4   3.361  0.00086 ***
## temp:seasonSummer   2488.4      990.0   2.513  0.01240 *
## temp:seasonFall    -3512.5     1337.8  -2.626  0.00902 **
## temp:seasonWinter    172.8     1100.6   0.157  0.87531
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 745.7 on 357 degrees of freedom
## Multiple R-squared:  0.7131, Adjusted R-squared:  0.7075
## F-statistic: 126.8 on 7 and 357 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.71, which is better than both the temp-only and season-only models. The interaction terms are significant, indicating that the effect of temperature on rentals varies by season.

cnt ~ month*weathersit

```r
model_month_weathersit <- lm(cnt ~ mnth * weathersit, data = bike_2011)
summary(model_month_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ mnth * weathersit, data = bike_2011)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2538.00  -340.22    42.93   376.17  1569.71
##
## Coefficients: (5 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       1235.53     150.50   8.209 4.91e-15 ***
## mnth2                              616.58     207.17   2.976 0.003131 **
## mnth3                             1099.94     212.85   5.168 4.08e-07 ***
## mnth4                             2506.76     223.96  11.193  < 2e-16 ***
## mnth5                             3374.94     207.17  16.291  < 2e-16 ***
## mnth6                             3710.33     202.46  18.327  < 2e-16 ***
## mnth7                             3313.54     192.13  17.246  < 2e-16 ***
## mnth8                             3436.95     198.48  17.316  < 2e-16 ***
## mnth9                             3628.65     240.12  15.112  < 2e-16 ***
## mnth10                            3373.30     209.87  16.073  < 2e-16 ***
## mnth11                            2402.02     204.71  11.734  < 2e-16 ***
## mnth12                            1554.87     204.71   7.596 3.11e-13 ***
## weathersitMist/Cloudy               47.47     228.63   0.208 0.835646
## weathersitLight Rain/Snow         -729.53     638.54  -1.143 0.254064
## mnth2:weathersitMist/Cloudy       -452.35     339.60  -1.332 0.183760
## mnth3:weathersitMist/Cloudy       -546.77     327.13  -1.671 0.095574 .
## mnth4:weathersitMist/Cloudy      -1010.89     324.73  -3.113 0.002012 **
## mnth5:weathersitMist/Cloudy       -639.44     323.47  -1.977 0.048880 *
## mnth6:weathersitMist/Cloudy       -587.88     336.74  -1.746 0.081768 .
## mnth7:weathersitMist/Cloudy         32.46     403.49   0.080 0.935938
## mnth8:weathersitMist/Cloudy      -1066.95     342.27  -3.117 0.001984 **
## mnth9:weathersitMist/Cloudy       -701.40     333.69  -2.102 0.036304 *
```

```
## mnth10:weathersitMist/Cloudy     -1147.80     334.92  -3.427 0.000686 ***
## mnth11:weathersitMist/Cloudy      -366.88     355.72  -1.031 0.303118
## mnth12:weathersitMist/Cloudy       297.24     338.10   0.879 0.379956
## mnth2:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth3:weathersitLight Rain/Snow   -451.94     789.25  -0.573 0.567288
## mnth4:weathersitLight Rain/Snow  -2217.76     905.71  -2.449 0.014854 *
## mnth5:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth6:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth7:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth8:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth9:weathersitLight Rain/Snow  -1951.99     755.71  -2.583 0.010220 *
## mnth10:weathersitLight Rain/Snow -2056.97     746.65  -2.755 0.006192 **
## mnth11:weathersitLight Rain/Snow  -845.02     745.21  -1.134 0.257637
## mnth12:weathersitLight Rain/Snow  -411.37     787.10  -0.523 0.601570
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 620.5 on 334 degrees of freedom
## Multiple R-squared:  0.8141, Adjusted R-squared:  0.7974
## F-statistic: 48.76 on 30 and 334 DF,  p-value: < 2.2e-16
```

This model produces the best R2 so far of 0.814. All of the months are significant predictors of rentals, and some interaction terms are also significant, indicating that the effect of month on rentals varies by weather situation. This makes sense as the impact of weather on rentals is likely to be more pronounced in certain months (e.g cloudy weather may have higher effect in summer months when it's usually hot).

4. Using Backward Selection

```r
# from part 3, we use the interaction plus all the predicts
regfit_backward <- regsubsets(cnt ~ season + mnth + holiday + weekday + workingday
                 + weathersit + temp + hum +
                   windspeed + temp:weathersit + temp:season +
                   mnth:weathersit,
                   data = bike_2011,
                 method="backward",
                 nvmax=50)
```

```
## Reordering variables and trying again:
```

```r
regfit_backward_summary <- summary(regfit_backward)

# Collect the metrics for each model
metrics <- data.frame(
  Model = 1:48,
  Backward_adjR2 = regfit_backward_summary$adjr2,
  Backward_cp = regfit_backward_summary$cp,
  Backward_bic = regfit_backward_summary$bic
)

metrics
```

```
##    Model Backward_adjR2 Backward_cp Backward_bic
## 1      1      0.5936550  668.026876    -317.9061
## 2      2      0.6554285  511.188108    -373.2019
## 3      3      0.6844826  437.613005    -400.4638
## 4      4      0.6867186  431.796166    -398.1723
## 5      5      0.7288677  326.048591    -446.0283
## 6      6      0.7640993  238.165677    -491.9530
## 7      7      0.7864723  182.799040    -523.4444
## 8      8      0.7995688  150.783544    -541.6711
## 9      9      0.8085009  129.264227    -553.4375
## 10    10      0.8175225  107.648272    -566.1808
## 11    11      0.8235916   93.429250    -573.6597
## 12    12      0.8294334   79.853164    -581.0869
## 13    13      0.8344176   68.458661    -587.0500
## 14    14      0.8390584   57.972050    -592.5675
## 15    15      0.8431101   48.984435    -597.0185
## 16    16      0.8457116   43.574028    -598.2691
## 17    17      0.8491558   36.159943    -601.6599
## 18    18      0.8523701   29.349127    -604.6750
## 19    19      0.8530594   28.660072    -601.5398
## 20    20      0.8538159   27.819527    -598.5834
## 21    21      0.8564953   22.388352    -600.4981
## 22    22      0.8587990   17.890733    -601.5710
## 23    23      0.8608949   13.919736    -602.1983
## 24    24      0.8626495   10.787416    -602.0036
## 25    25      0.8645041    7.443065    -602.1410
## 26    26      0.8661505    4.615787    -601.7814
## 27    27      0.8659387    6.179954    -596.3859
## 28    28      0.8662962    6.406686    -592.5454
## 29    29      0.8666680    6.605090    -588.7497
## 30    30      0.8670199    6.854806    -584.9058
## 31    31      0.8672900    7.299616    -580.8425
## 32    32      0.8674840    7.924409    -576.5744
## 33    33      0.8676871    8.531083    -572.3352
## 34    34      0.8677285    9.512779    -567.6538
## 35    35      0.8677858   10.458322    -563.0200
## 36    36      0.8681063   10.802664    -559.1170
## 37    37      0.8683203   11.394242    -554.9245
## 38    38      0.8682808   12.565531    -550.0329
## 39    39      0.8681363   13.974148    -544.8543
## 40    40      0.8679875   15.390623    -539.6675
## 41    41      0.8677052   17.105852    -534.1161
## 42    42      0.8674178   18.828571    -528.5558
## 43    43      0.8671290   20.550370    -522.9970
## 44    44      0.8668239   22.304513    -517.3989
## 45    45      0.8665138   24.065440    -511.7926
## 46    46      0.8661151   26.018702    -505.9501
## 47    47      0.8656980   28.007242    -500.0643
## 48    48      0.8652762   30.000000    -494.1733
```

```r
# Get model with best adjusted R2
best_model_adjR2 <- which.max(regfit_backward_summary$adjr2)
cat(sprintf("The best model according to adjusted R2 has %d predictors. \n", best_model_adjR2))
```

```
## The best model according to adjusted R2 has 37 predictors.
```

```r
cat(sprintf("Best R2: %.4f \n", regfit_backward_summary$adjr2[best_model_adjR2]))
```

```
## Best R2: 0.8683
```

```r
# View coefficients
coef(regfit_backward, best_model_adjR2)
```

```
##                       (Intercept)                     seasonSummer
##                        1598.84489                      -993.20047
##                        seasonFall                     seasonWinter
##                        1410.21139                      1371.16336
##                             mnth2                           mnth3
##                         459.61964                       993.12560
##                             mnth4                           mnth5
##                        1842.36058                      2307.95313
##                             mnth6                           mnth7
##                        2277.58917                      1715.27761
##                             mnth8                           mnth9
##                        1868.19989                      1978.09728
##                            mnth10                          mnth11
##                        1551.96614                       874.56997
##                            mnth12                     holidayYes
##                         553.21356                     -387.08766
##                        weekdayFri                     weekdaySat
##                          75.75023                       87.21857
##        weathersitLight Rain/Snow                      windspeed
##                       -1676.88163                    -1198.69988
##       weathersitMist/Cloudy:temp   weathersitLight Rain/Snow:temp
##                        -838.90953                    -1150.83477
##                 seasonSummer:temp      mnth3:weathersitMist/Cloudy
##                        3211.96450                     -243.31392
##      mnth4:weathersitMist/Cloudy      mnth7:weathersitMist/Cloudy
##                        -481.79133                      831.86664
##      mnth8:weathersitMist/Cloudy      mnth9:weathersitMist/Cloudy
##                        -424.49650                      -99.37622
##     mnth11:weathersitMist/Cloudy     mnth12:weathersitMist/Cloudy
##                         123.31298                      548.73769
##  mnth3:weathersitLight Rain/Snow  mnth4:weathersitLight Rain/Snow
##                         970.26863                     -542.81814
## mnth11:weathersitLight Rain/Snow mnth12:weathersitLight Rain/Snow
##                         559.29004                      604.14207
##  mnth5:weathersitLight Rain/Snow  mnth6:weathersitLight Rain/Snow
##                           0.00000                        0.00000
##  mnth7:weathersitLight Rain/Snow  mnth8:weathersitLight Rain/Snow
##                           0.00000                        0.00000
```

From the coefficients, we can see that the factors associated with more bike rentals are:
- Season (relative to Spring): Fall (+1410) and Winter (+1371).
- Months (relative to January): May (+2308), June (+2278), April (+1842), July (+1715), and September (+1978).

- Weekdays: Friday (+76) and Saturday (+87) - more retals near weekend.
- Higher temperature in summer: seasonSummer:temp (+3212).
Factors associated with fewer bike rentals are:
- Holiday (-387).
- Light Rain/Snow (-1677).
- High wind speed (-1199).
- Weather combined with temperature: Mist/Cloudy × Temperature (-839), Rain/Snow × Temperature (-1151).

5. Evaluate models on 2012 data

```
# Prepare data for 2012
bike_2012 <- bike_raw %>% filter(yr == 1)
```

```
# predict and print out the metrics
# Get coefficients
coef_backward <- coef(regfit_backward, best_model_adjR2)
# Build model matrix
X_2012 <- model.matrix(cnt ~ season + mnth + holiday + weekday + workingday +
                       weathersit + temp + hum + windspeed +
                       temp:weathersit + temp:season +
                       mnth:weathersit,
                       data = bike_2012)
# Generate prediction
vars <- names(coef_backward)

pred_2012 <- X_2012[, vars] %*% coef_backward
bike_2012$preds <- pred_2012

# metrics
backward_rsme <- sqrt(mean((bike_2012$cnt - bike_2012$preds)^2))
backward_r2   <- 1 - sum((bike_2012$cnt - bike_2012$preds)^2) / sum((bike_2012$cnt - mean(bike_2012$cnt

cat(sprintf("RMSE on 2012 data: %.2f \n", backward_rsme))
```

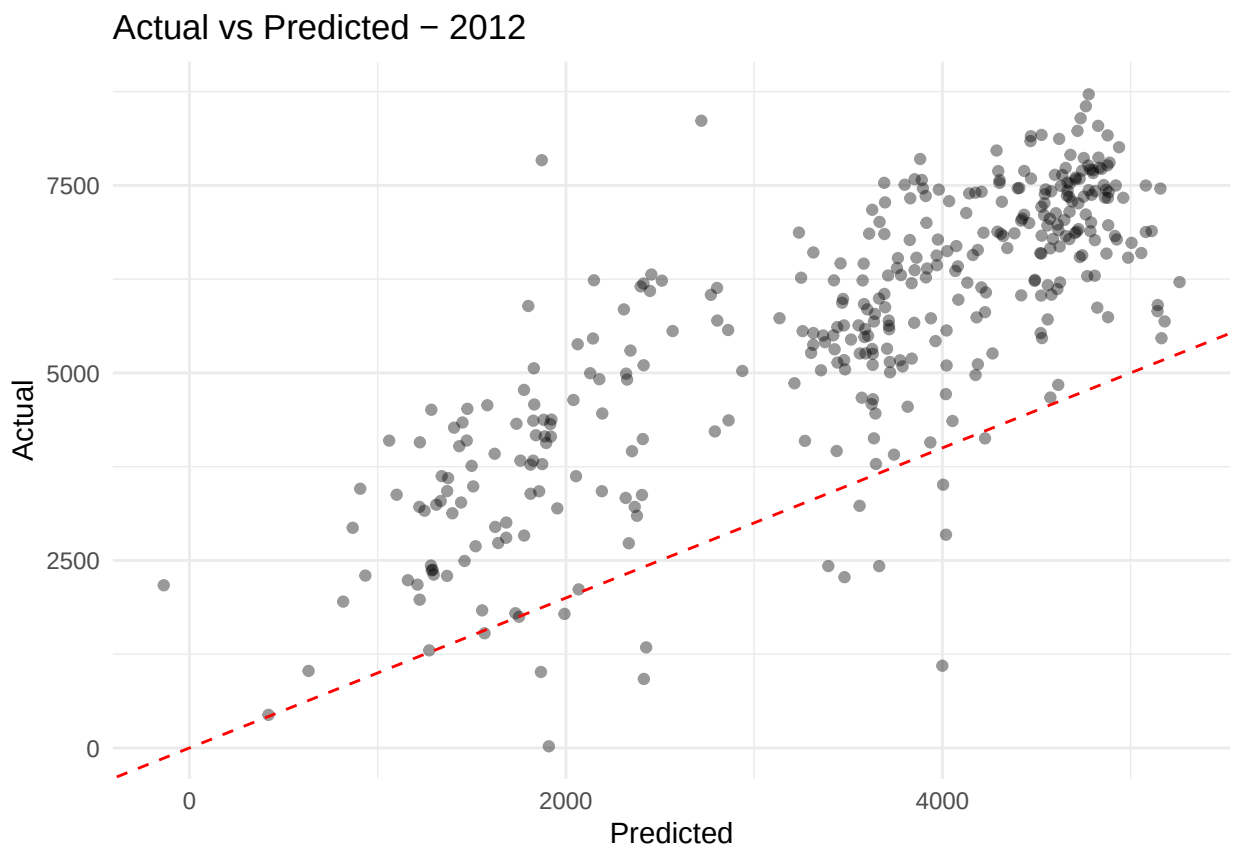
```
## RMSE on 2012 data: 2385.10
```

```
cat(sprintf("R2 on 2012 data: %.4f \n", backward_r2))
```

```
## R2 on 2012 data: -0.7830
```

-The model performs quite worse comparing to last part. This suggests us that model captured 2011 data pattern well but does not generalize perfectly to 2012.

```
# visualize the plot
ggplot(bike_2012, aes(x = preds, y = cnt)) + geom_point(alpha = 0.4) +
geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
labs(title = "Actual vs Predicted - 2012", x = "Predicted", y = "Actual") +
theme_minimal()
```

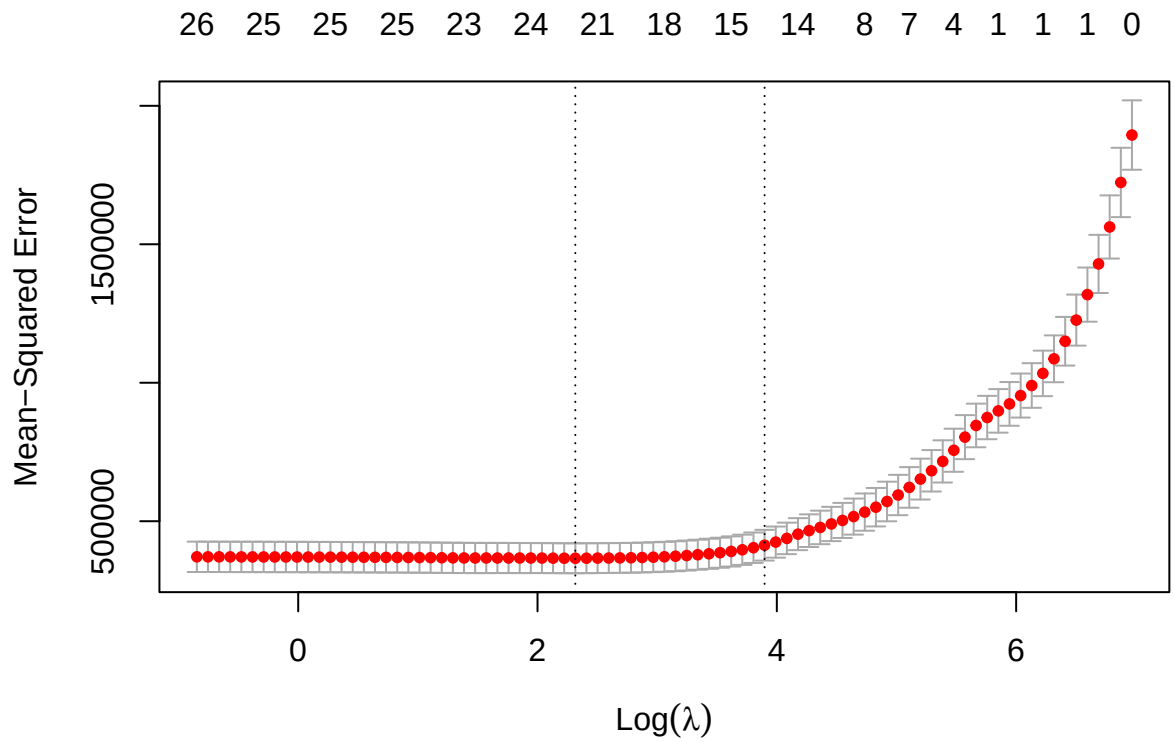


Part 2

1. Lasso, Ridge, PCR, PLS.

Lasso Regression

```
set.seed(1)
X_2011_matrix <- model.matrix(cnt ~ season + mnth + holiday + weekday + workingday +
                        weathersit + temp + hum + windspeed,
                        data = bike_2011)[, -1]
Y_2011 <- bike_2011$cnt
# Cross-validation to find optimal lambda
cv_lasso <- cv.glmnet(X_2011_matrix, Y_2011, alpha = 1)
plot(cv_lasso)
```




```r
# Best lambda
best_lambda <- cv_lasso$lambda.min
cat(sprintf("The best lambda based on cv is: %.2f \n", best_lambda))
```

```
## The best lambda based on cv is: 10.14
```

```r
# Train the lasso model with the best lambda
lasso_model <- glmnet(X_2011_matrix, Y_2011, alpha = 1, lambda = best_lambda)

# Get training r2
preds_2011_lasso <- predict(lasso_model, newx = X_2011_matrix)
r2_lasso_2011 <- 1 - sum((Y_2011 - preds_2011_lasso)^2) / sum((Y_2011 - mean(Y_2011))^2)
cat(sprintf("Training R2 for best lasso model: %.2f \n", r2_lasso_2011))
```

```
## Training R2 for best lasso model: 0.84
```

```r
cat(sprintf("Coefficients of the best lasso model: \n"))
```

```
## Coefficients of the best lasso model:
```

```r
print(coef(lasso_model))
```

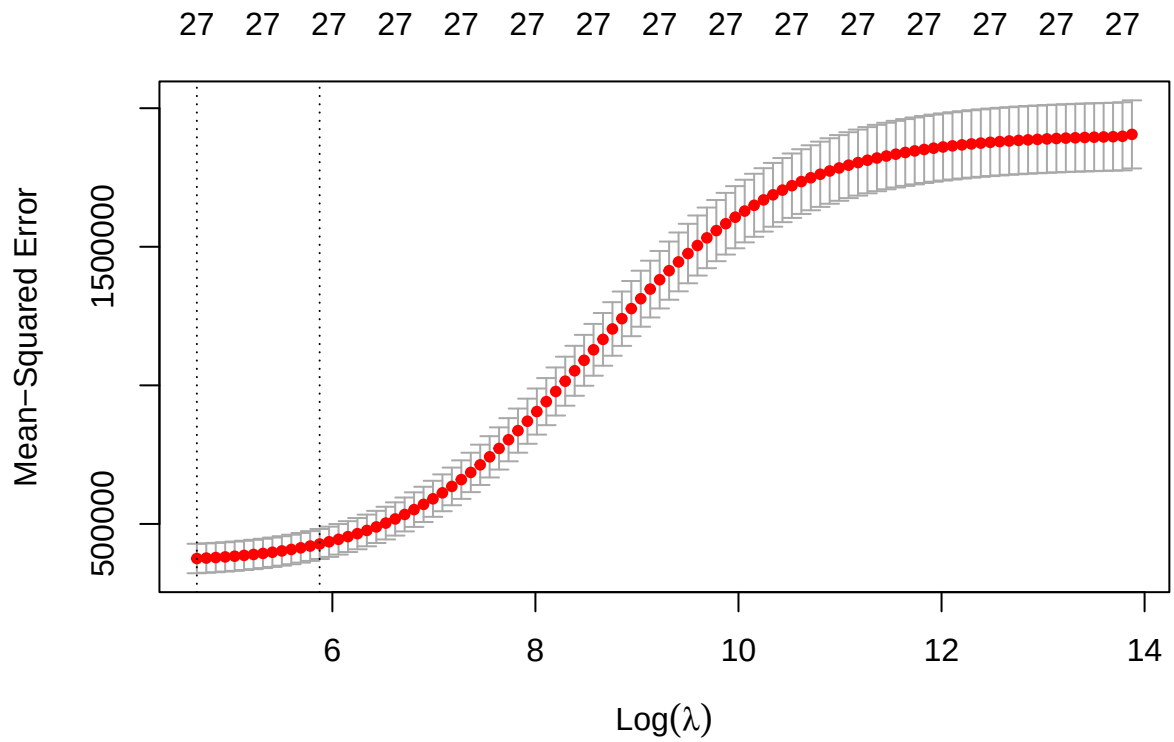
```
## 29 x 1 sparse Matrix of class "dgCMatrix"
```

```
##                                  s0
## (Intercept)                1848.12167
## seasonSummer                572.79329
## seasonFall                  941.16205
## seasonWinter               1304.73088
## mnth2                       -40.46750
## mnth3                         .
## mnth4                       291.69442
## mnth5                       956.79962
## mnth6                       611.70463
## mnth7                       -98.29208
## mnth8                         .
## mnth9                       446.76064
## mnth10                      354.15493
## mnth11                        .
## mnth12                       59.68473
## holidayYes                 -277.30527
## weekdayMon                    .
## weekdayTue                   16.81528
## weekdayWed                    .
## weekdayThu                    .
## weekdayFri                   47.35515
## weekdaySat                   80.01544
## workingdayYes                41.28544
## weathersitMist/Cloudy      -287.86914
## weathersitLight Rain/Snow -1585.07009
## weathersitHeavy Rain/Snow     .
## temp                       3826.68727
## hum                       -1153.32660
## windspeed                 -1889.47294
```

The lasso model that produce the best cv MSE has a lamda of 10.14. The training R2 for this model is 0.84 which is slightly lower than the model selected by backward selection (0.86). This model shrinks the coefficients of some predictors to 0, bringing the total number of predictor to 21.

Ridge Regression

```r
set.seed(1)
# Cross-validation to find optimal lambda
cv_ridge <- cv.glmnet(X_2011_matrix, Y_2011, alpha = 0)
plot(cv_ridge)
```




```r
# Best lambda
best_lambda_ridge <- cv_ridge$lambda.min
cat(sprintf("The best lambda for ridge regression based on cv is: %.2f \n", best_lambda_ridge))
```

```
## The best lambda for ridge regression based on cv is: 106.19
```

```r
# Train the ridge model with the best lambda
ridge_model <- glmnet(X_2011_matrix, Y_2011, alpha = 0, lambda = best_lambda_ridge)

# Get training r2
preds_2011_ridge <- predict(ridge_model, newx = X_2011_matrix)
r2_ridge_2011 <- 1 - sum((Y_2011 - preds_2011_ridge)^2) / sum((Y_2011 - mean(Y_2011))^2)
cat(sprintf("Training R2 for best ridge model: %.2f \n", r2_ridge_2011))
```

```
## Training R2 for best ridge model: 0.83
```

```r
cat(sprintf("Coefficients of the best ridge model: \n"))
```

```
## Coefficients of the best ridge model:
```

```r
print(coef(ridge_model))
```

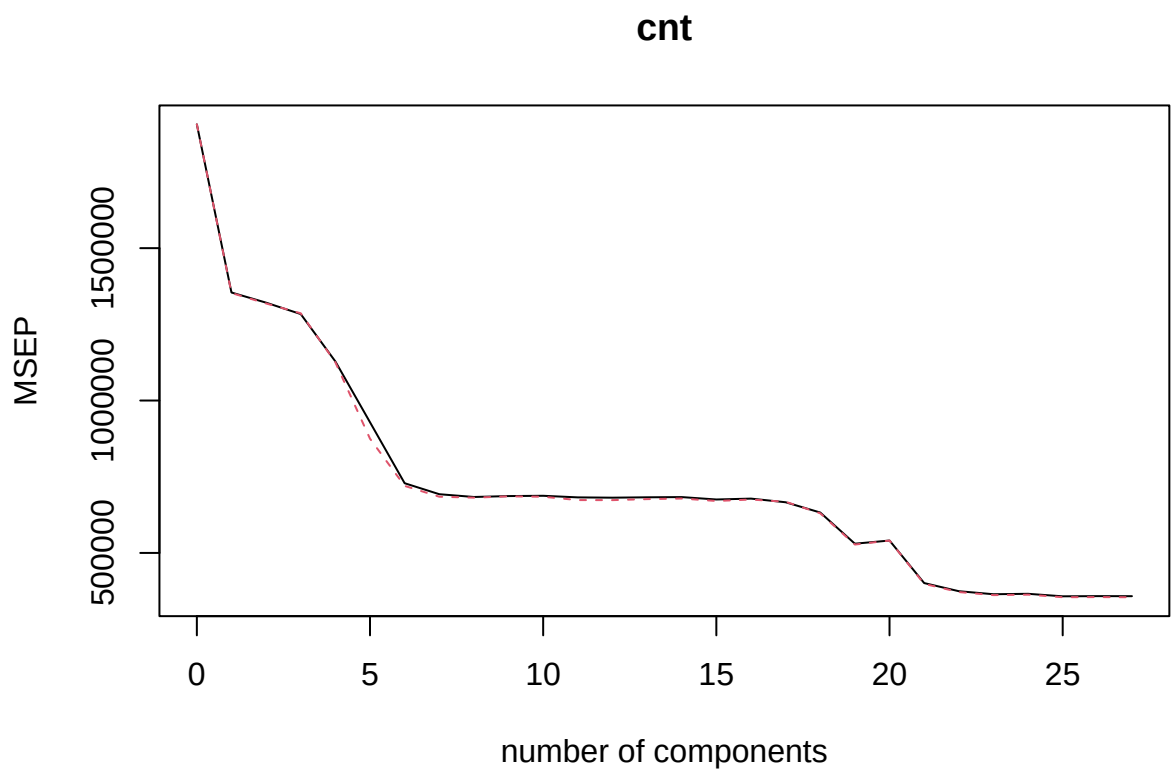
```
## 29 x 1 sparse Matrix of class "dgCMatrix"
```

```
##                                    s0
## (Intercept)                1997.6208888
## seasonSummer                481.5682290
## seasonFall                  712.1666191
## seasonWinter                943.4132434
## mnth2                      -146.8585459
## mnth3                       -24.0850214
## mnth4                       390.9599543
## mnth5                      1056.0361816
## mnth6                       839.1078298
## mnth7                       229.3104058
## mnth8                       350.1065163
## mnth9                       746.6363354
## mnth10                      690.3930687
## mnth11                      327.9448767
## mnth12                      233.1962164
## holidayYes                 -296.5414142
## weekdayMon                    0.1571474
## weekdayTue                   42.7630888
## weekdayWed                  -14.8382482
## weekdayThu                  -14.3307072
## weekdayFri                   61.3868174
## weekdaySat                  119.8698368
## workingdayYes                80.8910571
## weathersitMist/Cloudy      -299.1773039
## weathersitLight Rain/Snow -1526.9353360
## weathersitHeavy Rain/Snow     .
## temp                       3329.1828313
## hum                       -1059.9392846
## windspeed                 -1928.3625489
```

The ridge model that produce the best cv MSE has a lamda of 106.19. This model's training R2 is 0.83, which is the same as the lasso model. Both fits the data well, but the ridge model does not perform variable selection and thus retains all predictors with non-zero coefficients.

PCR

```r
set.seed(1)
X_2011_matrix <- X_2011_matrix[, apply(X_2011_matrix, 2, var) > 0]
prc_fit <- pcr(cnt ~ X_2011_matrix,
                       data = bike_2011, scale = TRUE, validation = "CV")
validationplot(prc_fit, val.type = "MSEP")
```




```
cat(sprintf("M with lowest MSEP for PCR: %d\n", which.min(prc_fit$validation$PRESS)))
```

```
## M with lowest MSEP for PCR: 25
```

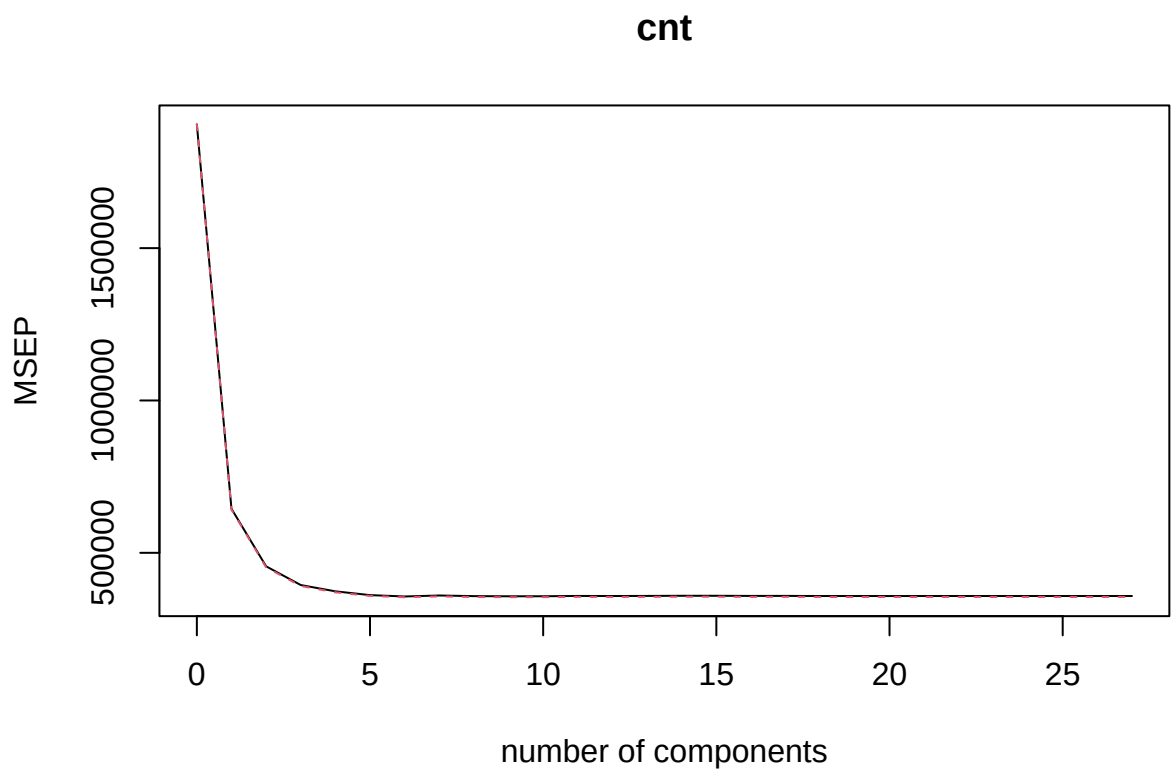
```
# Get model with best M and its training R2
best_M <- which.min(prc_fit$validation$PRESS)
preds_2011_pcr <- predict(prc_fit, ncomp = best_M)
r2_pcr_2011 <- 1 - sum((Y_2011 - preds_2011_pcr)^2) / sum((Y_2011 - mean(Y_2011))^2)
cat(sprintf("Training R2 for best PCR model: %.2f \n", r2_pcr_2011))
```

```
## Training R2 for best PCR model: 0.84
```

The PCR model that produces the best cv MSE has M = 25. The training R2 for this model is 0.84, which is on par with the lasso and ridge models.

PLS

```
set.seed(1)
pls_fit <- plsr(cnt ~ X_2011_matrix,
                       data = bike_2011, scale = TRUE, validation = "CV")
validationplot(pls_fit, val.type = "MSEP")
```




```r
cat(sprintf("M with lowest MSEP for PLS: %d\n", which.min(pls_fit$validation$PRESS)))
```

```
## M with lowest MSEP for PLS: 6
```

```r
# Get model with best M and its training R2
best_M_pls <- which.min(pls_fit$validation$PRESS)
preds_2011_pls <- predict(pls_fit, ncomp = best_M_pls)
r2_pls_2011 <- 1 - sum((Y_2011 - preds_2011_pls)^2) / sum((Y_2011 - mean(Y_2011))^2)
cat(sprintf("Training R2 for best PLS model: %.2f \n", r2_pls_2011))
```

```
## Training R2 for best PLS model: 0.84
```

The best PLS model has M = 6. The training R2 for this model is also 0.84.

2. Fit a GAM with some non-linear predictors.

From the exploratory analysis above, we can see that all quantitative predictors appear to have non-linear relationship with the target variable. Thus, we will fit these predictors with non-linear components. We will exclude atemp from the model due to its high collinearity with temp.

```r
# Fit a gam with non-linear components for quantitative variables
# (temp, hum, and windspeed; atemp excluded due to collinearity with temp)
gam_model <- gam(cnt ~ s(temp) + s(hum) + s(windspeed)
```

```r
                 + season + mnth + holiday + weekday + workingday + weathersit,
                 data = bike_2011)
summary(gam_model)
```

```
##
## Call: gam(formula = cnt ~ s(temp) + s(hum) + s(windspeed) + season +
##     mnth + holiday + weekday + workingday + weathersit, data = bike_2011)
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2232.61  -242.45    26.87   274.39  1696.42
##
## (Dispersion Parameter for gaussian family taken to be 233481.7)
##
##     Null Deviance: 691950048 on 364 degrees of freedom
## Residual Deviance: 76815414 on 328.9997 degrees of freedom
## AIC: 5583.637
##
## Number of Local Scoring Iterations: NA
##
## Anova for Parametric Effects
##               Df    Sum Sq   Mean Sq   F value    Pr(>F)
## s(temp)        1 408580841 408580841 1749.9478 < 2.2e-16 ***
## s(hum)         1  24612382  24612382  105.4146 < 2.2e-16 ***
## s(windspeed)   1  36655624  36655624  156.9957 < 2.2e-16 ***
## season         3  44405276  14801759   63.3958 < 2.2e-16 ***
## mnth          11  13989303   1271755    5.4469 5.568e-08 ***
## holiday        1   1506347   1506347    6.4517   0.01154 *
## weekday        6    968346    161391    0.6912   0.65686
## weathersit     2  14934176   7467088   31.9815 2.032e-13 ***
## Residuals    329  76815414    233482
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
##              Npar Df Npar F     Pr(F)
## (Intercept)
## s(temp)            3 37.096 < 2.2e-16 ***
## s(hum)             3  6.511 0.0002728 ***
## s(windspeed)       3  4.044 0.0076194 **
## season
## mnth
## holiday
## weekday
## workingday
## weathersit
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Calculate training R2
preds_2011_gam <- predict(gam_model)
r2_gam_2011 <- 1 - sum((Y_2011 - preds_2011_gam)^2) / sum((Y_2011 - mean(Y_2011))^2)
cat(sprintf("Training R2 for GAM model: %.2f \n", r2_gam_2011))
```

```
## Training R2 for GAM model: 0.89
```

The GAM model has a training R2 of 0.89, which is higher than all the previous models. Anova for Parametric Effects show that the linear components of season, month, holiday, weather situation, temp, hum, and windspeed are all significant. Anova for Non-parametric Effects show that the non-linear components of temp, hum, and windspeed are also significant. This suggests that the relationship between these predictors and total rentals is not strictly linear, and the GAM model captures this non-linearity to improve fit.

3. Evaluate the models on 2012 data.

```r
# Prepare 2012 data
X_2012_matrix <- model.matrix(cnt ~ season + mnth + holiday + weekday + workingday +
                       weathersit + temp + hum + windspeed,
                       data = bike_2012)[, -1]
Y_2012 <- bike_2012$cnt


# Perform prediction and calculate RSME and R2 for each model
# Lasso
preds_2012_lasso <- predict(lasso_model, newx = X_2012_matrix)
rmse_lasso <- sqrt(mean((Y_2012 - preds_2012_lasso)^2))
r2_lasso <- 1 - sum((Y_2012 - preds_2012_lasso)^2) / sum((Y_2012 - mean(Y_2012))^2)

# Ridge
preds_2012_ridge <- predict(ridge_model, newx = X_2012_matrix)
rmse_ridge <- sqrt(mean((Y_2012 - preds_2012_ridge)^2))
r2_ridge <- 1 - sum((Y_2012 - preds_2012_ridge)^2) / sum((Y_2012 - mean(Y_2012))^2)

# PCR
X_2012_matrix <- X_2012_matrix[, apply(X_2012_matrix, 2, var) > 0]
preds_2012_pcr <- predict(prc_fit, newdata = X_2012_matrix, ncomp = best_M)
rmse_pcr <- sqrt(mean((Y_2012 - preds_2012_pcr)^2))
r2_pcr <- 1 - sum((Y_2012 - preds_2012_pcr)^2) / sum((Y_2012 - mean(Y_2012))^2)

# PLS
preds_2012_pls <- predict(pls_fit, newdata = X_2012_matrix, ncomp = best_M_pls)
rmse_pls <- sqrt(mean((Y_2012 - preds_2012_pls)^2))
r2_pls <- 1 - sum((Y_2012 - preds_2012_pls)^2) / sum((Y_2012 - mean(Y_2012))^2)

# GAM
preds_2012_gam <- predict(gam_model, newdata = bike_2012)
rmse_gam <- sqrt(mean((Y_2012 - preds_2012_gam)^2))
r2_gam <- 1 - sum((Y_2012 - preds_2012_gam)^2) / sum((Y_2012 - mean(Y_2012))^2)

# Put results in a data frame for comparison
results <- data.frame(
  Model = c("Backward Selection", "Lasso", "Ridge", "PCR", "PLS", "GAM"),
  RMSE = c(backward_rsme, rmse_lasso, rmse_ridge, rmse_pcr, rmse_pls, rmse_gam),
  R2 = c(backward_r2, r2_lasso, r2_ridge, r2_pcr, r2_pls, r2_gam)
)

print(results)
```

```
##                 Model     RMSE         R2
## 1 Backward Selection 2385.101 -0.7829634
## 2               Lasso 2292.234 -0.6468223
## 3               Ridge 2314.331 -0.6787252
## 4                 PCR 2288.412 -0.6413357
## 5                 PLS 2279.998 -0.6292877
## 6                 GAM 2223.885 -0.5500783
```

The R2 values are all negative, indicating poor generalization. Because overall ridership grew from 2011 to 2012, models trained purely on 2011 data may systematically underpredict 2012 counts. The GAM model generally achieves the lowest RMSE and best R2 on the 2012 data due to its ability to capture non-linear relationships. The lasso, PCR, and PLS models show similar performance to each other. The backward selection model performs worst out-of-sample, overfitting to 2011-specific patterns that do not fully generalize to 2012.